

Responsible use of AI in Evidence Synthesis (RAISE 2026) 1: Recommendations for practice

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People who meet the ICMJE criteria for authorship have been listed as authors:

- *Substantial contributions to the conception or design of the work (via reviewing and editing as no primary research involved); or the acquisition, analysis, or interpretation of data for the work; AND*
- *Drafting the work or reviewing it critically for important intellectual content; AND*
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- *Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.*

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Abstract

The evidence synthesis field needs community-driven best practices to facilitate safe and effective AI use. Based on the principles of research integrity, the Responsible AI use in evidence Synthesis (RAISE) project provides recommendations and guidance to ensure the responsible, transparent and ethical use of AI in evidence synthesis.

RAISE 1 provides tailored recommendations for eight distinct roles in the evidence synthesis ecosystem: evidence synthesists, methodologists, AI tool development teams, organisations that produce evidence synthesis, publishers, funders, users, and trainers of evidence synthesis methods. The recommendations are interlinked; therefore, all roles need to collaborate to support others in adhering to them. It is important to note that one person or organisation may span multiple roles. RAISE aims to bridge roles and communities and align expectations and standards.

Evidence synthesis has been built on the principles of rigor, transparency and replicability, with an understanding that these serve as the cornerstones of evidence-based decision making. However, satisfying these requirements is time-consuming and expensive, and makes it difficult to keep up with the demand of funders and users. While AI promises to improve timeliness and cost-effectiveness, it also presents challenges. At the time of writing, many AI solutions would undermine these principles, risking a deluge of unreliable, biased and low-quality systematic reviews and syntheses. As AI becomes embedded within evidence synthesis processes, RAISE encourages all members of the community to adopt responsible AI practices to ensure that the core values on which the field was built remain intact as it evolves.

Highlights

What is already known

- There is considerable excitement about the potential for AI to make the evidence synthesis process more efficient, but also noticeable alarm over the potential for harm
- Guidance is needed to help evidence synthesis authors decide when (and how) a given tool might be utilised, and when it should not be

What is new

- A lack of robust and generalisable evidence makes it difficult to recommend the general adoption of tools across the field
- We envision an ecosystem of roles (including tool developers, evidence synthesists, and methodologists) working collectively to build tools and an evidence base supporting reliable use of AI in evidence synthesis
- The three RAISE papers cover recommendations specific for each role; guidance on development, evaluation, and reporting of tools; and guidance to support decisions about how and when tools might be adopted, and how to report tool use

Potential impact for RSM readers

- Should the field develop in accordance with the guidance, it should be able to take advantage of the benefits of AI, whilst avoiding the dilution of standards

Introduction

Why is this guidance needed?

Evidence syntheses, including systematic reviews, rapid reviews, scoping reviews, and other forms of evidence syntheses, are expected to accurately represent the available, relevant evidence to inform policy and practice decision. However, this work is often time-consuming and expensive due to its comprehensive and rigorous nature (i.e., identifying as much relevant research as possible, while meeting high standards of research integrity and quality). Further, evidence synthesis is a mostly manual process, requiring significant resources to meet these standards. This has often resulted in evidence synthesis ecosystems in which the best available evidence does not always inform policy and practice decisions.¹

AI use in evidence synthesis has rapidly evolved in recent decades as overviewed in the *current state of AI tools* in RAISE 3. Since the 2000s, machine learning and other types of automation advanced as potential solutions to replace time-consuming, manual tasks and thereby increase the efficiency and pace of evidence synthesis practices.² Significant progress in some areas has been made³, including the use of machine learning to identify randomized trials since 2016 by Cochrane⁴. Yet many evidence synthesis workflows remain mostly manual, expensive, and slow.ⁱ

Developments in artificial intelligence (AI – please see below), particularly large language models (LLMs) that can ‘understand’ content without the need for additional expensive training data from those using the systems, are rapidly changing the landscape. The explosion of research and use of AI-based tools outpaces that of previous technologies.⁵ However, there are concerns that many of these tools are ‘black boxes’ lacking generalisability, explainability, and transparency in assumptions, biases and limitations.

Research integrity is an important ethical framework that underpins good evidence synthesis. The development and use of AI in evidence synthesis should not compromise these established standards, or the quality or reliability of the findings or conclusions. Thus, there are two principal reasons for developing this guidance.

The evidence synthesis field needs to coordinate community-based best practices that enable better use of automation and AI in evidence synthesis. To our knowledge, at the time of writing, there is no comprehensive effort to coordinate AI development, evaluation/validation, documentation, and adoption across interested evidence synthesis parties including for:

- **Evidence synthesists** to standardise workflows and gain efficiencies by using AI;
- **Methodologists** to improve the evidence base for AI use in synthesis methods and update existing best practice;
- **AI tool developers** to design, develop, evaluate and disseminate tools, including developing and writing prompts for use in evidence synthesis;
- **Organisations that produce evidence synthesis** who can use it to inform their strategies for implementing and mainstreaming AI use across their communities.
- **Funders and commissioners of evidence synthesis** to design and evaluate programs for future investment;

- **Publishers of evidence synthesis** to enforce policies and publication practices that aim to improve the quality, reliability, and transparency of the published scientific literature;
- **Trainers of evidence synthesis methods** to enforce best practices that ensure AI use is shared and taught in a way that helps maximise efficiency while maintaining scientific integrity; and
- **Users** to understand and interpret AI use in evidence synthesis to ensure the transparent and responsible application of the findings.

Thus, a framework is needed within the ‘evidence synthesis ecosystem’ to define the considerations for AI use for each of these roles, as well as guidance to ensure consistent application and facilitate transparent, reliable, and unbiased development, evaluation, and adoption of AI in evidence synthesis.

Developing the recommendations

We established an international team of individuals with an interest and expertise in developing expert-based recommendations for responsible AI use in evidence synthesis. The team was led by individuals from the International Collaboration for Automation in Systematic Reviews (ICASR), Cochrane, and Campbell (JT, EF, ANS, WM), and includes interest holders from across public, private, government, and academic sectors of the evidence synthesis field.

Step 1

We used an online survey to raise awareness about the RAISE project and its aims, and gauge interest from individuals from across the field who wanted to contribute (open May to August 2024; 164 respondents). Team leads also raised awareness through their networks and the networks of their associated organisations.

Step 2

With the pillars of research integrity as the foundation for the project,⁶ we developed recommendations for the responsible use of AI in evidence synthesis for different roles within the evidence synthesis ecosystem based on developing expectations from across research, publishing and legal and regulatory frameworks.^{7–11} This was posted [on the Open Science Framework](#) in September 2024, and an online survey was used to gather feedback (43 respondents by end of 2024). Feedback was requested on the areas of expertise for each role in the evidence synthesis ecosystem, the level of agreement with the proposed recommendations, and requests for other developments and evidence from across the field to inform the recommendations (see RAISE project files ‘2024_RAISE 1 feedback survey’ for the survey questions and high-level survey responses). To make the survey manageable, individuals were asked only about the expertise and recommendations for roles they self-identified with. Promotional materials for the surveys were shared with the networks of the people and organisations associated with RAISE to promote via email and social media, as well as through presentations from members of the RAISE team. Additional feedback was received via email.

Informed consent to participate was assumed when participants submitted survey responses, given that participation was optional. The authors acknowledge that this survey-based approach has important limitations that should be considered when interpreting these recommendations.

Recruitment through the networks of RAISE team members introduces a risk of self-selection bias, in that respondents are likely to hold pre-existing views sympathetic to responsible AI use and may not represent the range of opinion across the full evidence synthesis community.

Step 3

Extensive feedback received via the survey's open-text questions and email resulted in revisions to the recommendations and the development of more detailed guidance to support their implementation. The main revisions to the *Recommendations* included the addition of two roles ('users of evidence synthesis' and 'trainers of evidence synthesis methods'), as well as the creation of the second and third papers of the three-paper series on the Responsible AI in Evidence Synthesis (RAISE).

This paper, RAISE 1, includes *Recommendations for Practice* for the main roles in the evidence synthesis ecosystem to enhance collaboration and communication, thereby enabling the transparent and reliable use of AI in evidence synthesis. RAISE 2 contains guidance on *building and evaluating AI evidence synthesis tools*, focusing on determining whether an AI tool does what it claims to do to an acceptable standard. RAISE 2 includes guidance on how to build and validate AI tools, conduct evaluations to build a cumulative evidence base, and report evaluations, including performance metrics to consider. RAISE 3 offers guidance on *selecting and using AI evidence synthesis tools*, focusing on understanding whether an AI tool can be used for a specific evidence synthesis, including how to assess, select and use an AI tool. RAISE 3 includes ethical, legal and regulatory considerations, and the current state of AI tools for evidence synthesis.

We aim to ensure RAISE is living and evolves as the field develops. Future aims involve developing these recommendations and accompanying guidance documents into consensus-based guidelines for responsible AI use in evidence synthesis. Future iterations of RAISE will explicitly consider global equity. The roles described across the RAISE ecosystem, i.e., tool developers, funders, methodological organisations, publishers, are predominantly concentrated in high-income countries, as is much of the evidence base on which RAISE draws. The recommendations assume institutional computing infrastructure, commercial AI subscriptions, and organisational governance capacity that cannot be taken for granted in low- and middle-income country (LMIC) settings, where evidence synthesis may rely on individual researchers or small, under-resourced teams. RAISE will therefore: (i) identify which recommendations are applicable regardless of resource level and which require LMIC adaptation; (ii) actively engage LMIC organisations in developing future versions; and (iii) advocate for funder investment in AI capacity-building as a prerequisite for equitable implementation.

What types of AI are covered by RAISE?

Although numerous definitions of AI exist across various domains, for the purpose of this guidance, AI is defined as follows:

... a set of advanced technologies that enable machines to do highly complex tasks effectively – which would require intelligence if a person were to perform them¹

¹ See Harwich and Laycock (2018, p.11) at http://www.reform.uk/wp-content/uploads/2018/01/AI-in-Healthcare-report_.pdf. See also: <https://oecd.ai/en/wonk/ai-system-definition-update>

This ranges from the application of:

- machine learning (ML) that uses algorithms that learn from data to make predictions;
- natural language processing (NLP) that understands and generates human language;
- expert systems that use rule-based algorithms to mimic human decisions;
- deep learning, which includes LLMs that focus on both text and language; and
- cross-cutting generative AI that might combine NLP and LLM to generate text, figures, or code.

The authors acknowledge that this definition is deliberately broad. It encompasses a wide range of tools with substantially different risk profiles, capabilities, and validation requirements. For practical application, readers should interpret this definition in conjunction with the technology taxonomy provided in RAISE 2 and the task-specific tool recommendations in RAISE 3, which provide the operational specificity that the definition itself does not. Where the term 'AI' appears in individual recommendations, it should be understood to refer to the relevant class or classes of technology described in those companion documents, not to all AI systems indiscriminately.

For an overview of the landscape at the time of writing, refer to *current state of AI tools* in RAISE 3, which includes active ML to screen literature search results; supervised and unsupervised clustering NLP to classify and categorize information;¹² data extraction through semi-automation based on ML;^{13,14} and risk of bias and report writing facilitated by generative AI.

What is responsible AI use in evidence synthesis?

Responsible AI is a framework for developing and using AI that does not compromise the principles of research integrity, which include honesty, rigour, transparency and open communication, care and respect, and accountability⁶. It also involves ensuring ethical, moral and legal values are upheld.^{7,8}

Considerable effort has gone into articulating what integrity in research is at a high level. However, precisely what this entails in practice must be specified on a field-by-field basis. It is essential that standards for responsible AI use are developed with the values of all key interest holders, so this paper explicitly identifies these groups (or roles) to provide role-specific recommendations and to facilitate discussion and community input.

An ecosystem of people and organisations

The evidence synthesis ecosystem comprises of individuals, collaborations, and organisations that play roles in developing and using AI responsibly (Figure 1). RAISE includes eight distinct roles, each with different expertise and needs, along with recommendations to define responsible AI use. Since these recommendations are interlinked, all roles need to collaborate across the ecosystem to support adherence to them.

Evidence synthesis methodologists working with AI developers will help to create tools that are fit for purpose, while evidence synthesists sharing data will enable validation within real-world contexts. Methodologists will test tools to establish best-practice guidance, and organisations will embed standards, monitor progress and support mainstreaming of tools. Trainers will work with methodologists to produce resources to support AI literacy, and to train evidence synthesists, AI developers and others to use and evaluate tools effectively. Publishers will enforce best practice by

publishing evidence syntheses that adhere to the guidance, and end users will be equipped with the knowledge and skills to assess appropriate AI use. Funders will support tools, infrastructure and methods development, and help the whole community to transition to using AI in evidence synthesis responsibly. However, this theory of change depends on several enabling conditions. AI developers must engage with evidence synthesis methodology and accept constraints driven by research integrity, not commercial considerations alone. Publishers and funders must enforce compliance with RAISE recommendations as a condition of publication or funding, rather than treating them as aspirational. The community must invest in training infrastructure sufficient to build AI literacy equitably across institutions. Methodologists must establish minimum empirical benchmarks for tool validation (see RAISE 3), such as recall thresholds for screening or inter-rater reliability minima for data extraction, so that guidance can be operationalised and audited.

With each role in the ecosystem playing its part, a virtuous cycle is created where tools are developed that are fit-for-purpose and ready for use, making evidence synthesis faster, and cost-effective, and ultimately leading to better decisions and improved outcomes. It is important to note that one person or organisation may cross-cut roles within the evidence synthesis ecosystem. For example, evidence synthesis methodologists, responsible for developing best practice guidance, are likely to also be evidence synthesists, responsible for generating high quality evidence syntheses; and some organisations that produce evidence synthesis will also publish it.

Figure 1: Overview of the roles in the evidence synthesis ecosystem and their recommendations for responsible AI use in evidence synthesis



Scope of the paper

This paper outlines recommendations for the key roles in the evidence synthesis ecosystem that contribute to responsible AI use. The recommendations are applicable to evidence synthesis across any topic or domain. We also anticipate that the same principles will apply across different types of evidence synthesis, including systematic reviews, systematic evidence maps, rapid reviews, scoping reviews and gap maps, etc.

This paper describes the specific recommendations for each role in the ecosystem.

1. Evidence synthesists

Role

Evidence synthesists are responsible for conducting reliable evidence synthesis and are accountable for the accuracy and relevance of their work. This includes anyone who is an author on the final evidence synthesis report, including general evidence synthesists, specialists in certain specific methods, such as information specialists or statisticians, and domain experts. They are responsible for following evidence synthesis best practice guidance and policies, but are not necessarily responsible for carrying out the evaluation or validation of AI tools.

Expertise

- Methods for planning, conducting and reporting evidence synthesis
- Relevant substantive domains (e.g., health, social, environmental areas, epidemiology)
- Classification of research and the extraction of data from research papers (which can be useful for training / developing AI tools)
- Accessing and interacting with users of evidence synthesis

However, they are not expected to have expertise in the technical details of how AI tools function. Indeed, evidence synthesis on a given topic are typically conducted by a team of subject matter experts, with varying degrees of experience or understanding of AI tools.

What they need to know

An evidence synthesist needs to know which tools are available, whether they have been validated, and how to use these tools in ways that enhance rather than undermine their work. Refer to *selecting and using an AI tool* and *assessing the suitability of an AI tool for use in an evidence synthesis* in RAISE 3 for a step-by-step guide for evidence synthesists who may be considering the use of AI in their evidence synthesis.

Recommendations for evidence synthesists

Remain ultimately responsible for the evidence synthesis ¹¹:

- 1.1 Evidence synthesists are accountable for the content, methods and findings of their evidence synthesis, including the decision to use AI, how it is used, and its impact on the synthesis.
- 1.2 When considering using an AI system or tool, be critical of its evaluations (see *building and validating an AI tool* and *conducting evaluations to build a cumulative evidence base* in RAISE 2), to understand whether it does what it states at an adequate level, as well as its limitations, and whether it can be applied to the context of the specific synthesis.
- 1.3 The use of AI should be justified and should involve demonstrating that the tools are methodologically sound, that they do not undermine the trustworthiness or reliability of the synthesis or its conclusions, and that they are appropriate for the specific evidence synthesis.
- 1.4 AI should not receive credit for authorship of an evidence synthesis, although it is sometimes employed as a writing aid. Human authors are accountable for the proper use of AI in their evidence synthesis; an AI tool cannot satisfy all authorship criteria.⁹
- 1.5 AI should not be used to fabricate or falsify evidence synthesis data, results or findings.

- 1.6 Evidence synthesists should be aware of and account for the growing prevalence of research papers that are partially or fully written by AI.
- 1.7 Evidence synthesists should not treat a chatbot (e.g., ChatGPT) as a knowledge base for evidence synthesis tasks; that is, do not rely on it for factually accurate answers.

Report AI use in your evidence synthesis manuscript transparently:

- 1.8 Authors should declare when they have used AI if it makes or suggests judgements, such as, in relation to the eligibility of a study, critical appraisals (including risk of bias assessments), extraction of bibliographic, numerical or qualitative data about a study or its results, synthesis of data from two or more studies, assessments of the certainty of evidence (including GRADE domains or overall certainty ratings for an outcome or finding), drafting text that summarises the overall strength of evidence, related implications for decision making or research, or plain language summaries.
- 1.9 Adhere to established reporting standards for evidence synthesis, such as the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)¹⁵ or RepOrting standards for Systematic Evidence Syntheses (ROSES).¹⁶ PRISMA, for example, includes items on reporting automation tools used at different stages of the synthesis process. In general, evidence synthesists should report:
 - a. the name(s) of the AI system(s)/tool(s)/platform(s), version(s) and date(s) it was used or they were used;
 - b. the purpose of using the AI tool and specify which parts of the evidence synthesis process were impacted. (Note: Authors should cite or reference user guidance or report how AI was used, including any modifications that were applied); the justification for using the AI tool, including:
 - i. evidence that the tool is methodologically sound and is unlikely to undermine the trustworthiness or reliability of the synthesis or its conclusions (e.g., citing or referencing evaluations of its performance that detail the impact of errors, limitations and generalisability, and),
 - ii. how it has been validated (and piloted, if applicable) to ensure that it is appropriate for use in the context of the specific evidence synthesis. Wherever possible and practical, make the inputs (e.g., prompt development), outputs, datasets, and code publicly and freely available (for instance, on repositories or as supplementary material], and ¹¹ description of any steps taken to review and verify AI-generated content;
 - c. any financial or non-financial interests in the AI tool, if applicable, alongside the funding sources of the AI tool, and;
 - d. discuss any limitations of using AI in the review processes, including any potential biases, and comment on the potential impact of limitations, where relevant.

Ensure ethical, legal and regulatory standards are adhered to when using AI: ¹¹

- 1.10 Ensure ethical, legal, and regulatory standards are upheld in the application of AI to your synthesis. For instance, consider issues related to plagiarism, provenance, copyright, intellectual property, jurisdiction, licensing, confidentiality, compliance, and privacy

responsibilities, including data protection laws. The need to stay up to date on technologies available and utilize cost-effective technology is important but should not come at the expense of responsible use. (see *ethical, legal and regulatory considerations when using an AI tool* in RAISE 3).

Contribute to the ecosystem to help all roles continue to develop and grow, for example:

- 1.11 Collaborate with those across the ecosystem to explore and define standards and expectations that can be applied globally.
- 1.12 Engage in training and commit to ongoing learning about the responsible use of AI as it evolves, and stay informed about best practices.
- 1.13 Share your feedback on your experience with AI with the developer, particularly in situations where the model did not perform as expected, if biases are detected, or if you notice contexts in which the system is less dependable.
- 1.14 Where possible, share data to allow AI system developers to test and validate their prototypes with real-world data, helping create reliable tools.
- 1.15 Participate in networks to learn and to share knowledge about responsible AI use in evidence synthesis.

2. Evidence synthesis methodologist

Role

Evidence synthesis methodologists are responsible for defining best practices and ensuring they facilitate reliable and trustworthy evidence synthesis. In the context of the use of AI, this work involves conducting evaluation and validation studies (including simulation studies) and integrating their findings into best-practice guidance.

Expertise

- Evidence synthesis methodology
- Methods for evaluating and validating AI tools for specific tasks
- AI system functionality, to appraise and critique their suitability

What they need to know

An evidence synthesis methodologist may not possess the in-depth knowledge of AI technologies that a computer scientist has, nor may they have programming expertise. However, they should have a solid understanding of how these tools function to build a robust evidence base.

Recommendations for evidence synthesis methodologists

Adhere to open science practices when researching and evaluating AI systems:

- 2.1 When conducting evaluations, recommendations for methodologists include:
 - a. Adhere to best practice when training or testing the application of AI within an evidence synthesis (see *conducting evaluations to build a cumulative evidence base* in RAISE 2).
 - b. Use appropriate comparators (e.g., current best practice, a human as the gold standard, another AI system that has previously been evaluated against current

- best practice). Some metrics, such as user experience, may not need a comparator.
- c. Use the most appropriate study design to address your research question, e.g., compare AI systems to non-AI processes across large numbers of evidence syntheses; conduct a single Study Within a Review¹⁷ comparing AI systems to non-AI processes; or conduct a feasibility evaluation to test whether an AI system is applicable to the context of a specific evidence synthesis, etc.
- 2.2 When validating AI tools, methodologists should report any equity issues or biases within the AI systems, including potential biases in the systems' training data and strategies for users to help mitigate possible bias-related issues (see *building and validating an AI tool* in RAISE 2 and *ethical, legal and regulatory considerations when using an AI tool* in RAISE 3).
 - 2.3 If you conduct a Study Within a Review, register the study and use template protocols from the Study Within a Review Network, if possible. This allows the findings from your single Study Within a Review to be combined with others, thereby building a cumulative evidence base.
 - 2.4 Evaluations should be conducted using appropriate methods (see *conducting evaluations to build a cumulative evidence base* in RAISE 2), including using key performance metrics (see *performance metric considerations for AI tool evaluations* in RAISE 2), adhering to protocols, judiciously interpreting data, and drawing conclusions from the findings with a clear explanation of the basis of the conclusions.
 - 2.5 Evaluations should be made publicly available, ideally published in peer-reviewed journals (although publishing as a preprint can also be beneficial, given the rapid developments in the field). They may also be uploaded to a repository or included as supplementary material for an evidence synthesis manuscript.
 - 2.6 Ensure full and transparent reporting of evaluations, including full methods and publishing of null results. Full reporting should also include:
 - a. the name of the AI system(s)/tool(s)/platform(s), version(s) and date(s) it was used or they were used;
 - b. how and where the data were collected (as the scope or source of the data may result in bias in the evaluation);
 - c. any exclusion criteria that were applied (as this may result in bias in the evaluation);
 - d. how data were labelled or annotated as part of building the model;
 - e. any quality assurance/control processes used to clean the data, including handling missing data and outliers; and
 - f. prompt engineering and the documentation of how prompts were developed and validated (for AI that require prompts, such as LLMs).
 - 2.7 Whenever possible, openly share datasets, source code, and other materials used to develop and run the evaluation(s), or clarify how they can be accessed, including any restrictions on their use or licensing considerations.

Commit to objective and impartial evaluations and validation of AI systems:

- 2.8 If methodologists have a financial interest in the tool being evaluated, this should be declared. While making every effort to ensure that the research is objective, the

evaluation or validation should not be presented as independent. It is also good practice to declare any non-financial interests.

Contribute to the ecosystem to help all roles continue to develop and grow, for example:

- 2.9 Collaborate with those across the ecosystem to explore and define standards and expectations that can be applied globally.
- 2.10 Engage in training and commit to ongoing learning about the responsible use of AI as it evolves, and stay informed about best practices.
- 2.11 Collaborate with AI developers to communicate the expected standards for responsible AI use in evidence synthesis. When implementing an AI system within evidence synthesis processes, follow responsible handover guidelines with the AI developer.¹⁸
- 2.12 Inform evidence synthesists, organisations that produce evidence synthesis, and AI developers of your findings on applicability, limitations, and generalisability, along with any additional recommendations and feedback.
- 2.13 Participate in networks to learn and to share knowledge about responsible AI use in evidence synthesis.

3. AI tool development teams

Role

AI tool development teams vary in size, including autonomous groups of people who are responsible for developing and deploying AI tools to assist with one or more evidence synthesis tasks. These teams include individuals who write the software code, such as software developers and computer scientists. They also include designers, product owners, quality assurance testers, and those involved in the marketing and sales of deployed products.

Expertise

- The technical details of how AI technologies work, including their strengths and weaknesses
- The needs of evidence synthesists and the principles of integrity that underpin their methodology and ensure their reliability²
- Building, deploying and supporting reliable software applications and application programming interfaces (APIs)

What they need to know

An AI tool development team should understand the principles that underpin reliable evidence synthesis and how they are applied in practice (see the footnote above). Teams should also know how to evaluate and validate tools and the expected/required levels of accuracy (see RAISE 2), though evaluation activities are also the responsibility of evidence synthesis methodologists.

² Note that every member of the team does not need this knowledge, but the product manager (or person in a similar role) should do, and there should be procedures in place to ensure that underpinning principles of evidence synthesis are not inadvertently broken during the software development process.

Recommendations for AI tool development teams

Evidence synthesis is a demanding use case for automation, requiring high accuracy and accountability.

Many guidelines emphasise the importance of AI having ‘human oversight’ or being ‘human-centred’.¹⁹ In this context, it means that humans should maintain control over the AI system, possessing agency and oversight of its operation within the method and/or workflow. The AI system should enhance human capabilities, rather than replace, with development teams collaborating with users to create solutions that meet operational needs. Additionally, human rights and democratic values, including non-discrimination and privacy, should be integrated within the AI.⁸

The following recommendations seek to promote responsible AI development and usage by teams involved in AI tool development. Recommendations for AI tool developers also support adherence to the principles of open scholarly infrastructure⁸ and follow Sense About Science’s responsible handover of AI framework¹⁸.

Adhere to open science practices when designing, building, testing and validating tools:

- 3.1 Ensure the development team appreciates the principles that underpin reliable evidence synthesis (see *building and validating an AI tool* in RAISE 2) and design AI that supports these principles and does not undermine them.
- 3.2 Use high-quality, appropriately sized datasets to develop AI systems and ensure no overlap or contamination between the datasets used for training, testing, and validation. If this is not feasible, report the limitations, assumptions, and the potential for overestimated performance.
- 3.3 Minimise bias in the data used to train systems as well as the system parameters and requirements.
- 3.4 Conduct training, testing, and validation evaluations using appropriate methods, including key performance metrics, adhering to protocols, interpreting data judiciously, and drawing reasonable conclusions based on the findings (see RAISE 2).
- 3.5 Ensure complete, transparent, and public reporting of training, testing, and validation evaluations, including training sets, detailed methods and the publication of null results, preferably in peer-reviewed journals (see *reporting the building and evaluation of an AI tool* in RAISE 2).
- 3.6 Where possible, openly share training, testing, and validation datasets, and provide details about the nature, extent, and context of the data. Transparency can be maximised if the source code and other materials used to develop and run the models are also openly available. These should ideally adhere to the FAIR (Findability, Accessibility, Interoperability, and Reuse) principles for data²⁰ and to the FAIR4RS principles for research software.²¹ Additionally, confirm how it can be accessed, along with any restrictions on its use or licensing considerations.
- 3.7 Awareness and adherence to national and international guidelines, such as the EU AI Act, especially regarding the technical information about models you have built or fine-tuned.²²

Be transparent about when the AI works best, its limitations and any interests:

See AI tool assessment considerations in RAISE 3 for an overview of what evidence synthesists need to know about an AI system to understand if they could use it in their synthesis, which includes the following.

- 3.8 Publicly and clearly report details about the system, including name, version and purpose. Include information about how the model has developed over time and what future updates are planned.
- 3.9 Report how the system works; do not hide reasonable openness behind ‘commercial confidentiality’ where this is unnecessary.
- 3.10 Be clear about which data are used to train, evaluate, and validate the tool, and where such data are stored and shared.
- 3.11 Provide clear guidance to users and help them take accountability for their use of your AI system. Comprehensive user guidance enhances transparency, facilitates accountability, promotes appropriate and ethical use, and ultimately helps evidence synthesists make informed decisions about their usage.
- 3.12 Report the strengths and limitations of the AI system(s), including what scenarios it is best suited to and when it may be inappropriate to use, with enough detail that users can decide whether it will be appropriate to use in their own work.
- 3.13 Report any equity issues or biases within the AI systems, including known or predicted limitations in generalisability based on the scope of the training and testing datasets or any biases that may exist in the systems’ training data and strategies for users to help mitigate potential issues arising from these biases (see *building and validating an AI tool* in RAISE 2).
- 3.14 Be transparent about your funding sources and declare any financial and non-financial interests.
- 3.15 Be honest about all aspects of AI system use and development.
- 3.16 Do not market or otherwise recommend a tool for use in evidence syntheses if you cannot demonstrate robust evaluation(s) that support the recommendation.
- 3.17 Tools which do not yet have robust evaluations (see *building and validating an AI tool* and *conducting evaluations to build a cumulative evidence base* in RAISE 2) demonstrating their fitness-for-purpose can be released, but user interfaces and documentation should make clear that they are released for testing and evaluation only.

Commit to continued learning, development and monitoring of the AI system:

- 3.18 As stated in the guidance for the EU AI Act, “...assess and mitigate systemic risks, particularly by conducting model evaluations, tracking and documenting serious incidents, and ensuring adequate cybersecurity protection for both the model and its physical infrastructure.”²²
- 3.19 Respect the accuracy standards expected by the evidence synthesis community. Avoid promoting systems that undermine these principles. For instance, do not develop, promote, or deploy systems that may amplify publication bias or generate biased and misleading summaries (see *Building and Validating an AI Tool* in RAISE 2).
- 3.20 Facilitate the use and development of human-centred AI systems that emphasise human oversight. In this context, ‘human-centred’ means that humans should maintain control over the AI system while having agency and oversight of its operation.

- 3.21 Have an accessible and transparent appeals or feedback pathways for when systems do not perform as expected, especially those that may highlight biases in the training dataset or contexts in which the system is less reliable.
- 3.22 Continuously monitor deployed systems to ensure that performance does not change unexpectedly over time.

Contribute to the ecosystem to help all roles continue to develop and grow, for example:

- 3.23 Collaborate with those across the ecosystem to explore and define standards and expectations that can be applied globally.
- 3.24 Engage in training and commit to ongoing learning about the responsible use AI as it evolves, and make sure you remain informed about best practices.
- 3.25 Engage and involve key interest holders in the development of AI systems or data governance frameworks to ensure that the necessary standards and expectations for responsible use are met.
- 3.26 When transitioning an AI system from development to adoption, independent of whether this is through embedding an AI system within a tool or platform for general use or applying an AI system within specific evidence synthesis, follow responsible handover guidelines (such as those provided by Sense About Science ¹⁸).
- 3.27 Do not require users of your AI system to relinquish rights over or share their data for future system development and evaluation. Ensure your expectations align with existing and emerging standards for regulation and use.
- 3.28 Ensure that ethical, legal, and regulatory standards are followed during AI model development, such as avoiding copyright infringement or unlawful use, and clearly outline any ethical or legal considerations that users of your AI system should be aware of, such as those relating to privacy, copyright, and other relevant areas (see *ethical, legal and regulatory considerations when using an AI tool* in RAISE 3).
- 3.29 Consider taking steps to discourage the treatment of your AI system as an author and avoid anthropomorphising it and its outputs (for instance, refrain from giving it a human-like name).

4. Organisations that produce evidence synthesis

Role

Organisations that produce evidence synthesis are responsible for setting and implementing standards and policies for planning, conducting and reporting evidence synthesis across their organisation. They also collaborate with other organisations, evidence synthesists, and methodologists to develop standards that apply across the ecosystem. Therefore, they support the ecosystem by creating actionable standards and demonstrating their application in their evidence synthesis products and policies. Additionally, they may also be directly involved in the development of AI tools, and can facilitate access to synthesis data for AI tool development teams. They may also oversee the responsible technical implementation of AI, considering economic and environmental impacts as well as equitable access to the technology.

Expertise

- The development and implementation of standards
- Evidence synthesis best practices
- The needs of their user communities and of the field
- Organisational policies, procedures, and requirements

What they need to know

Organisations that produce evidence synthesis need to know what constitutes the best and most reliable practice in their domain and which tools and technologies are suitable and acceptable for use in their organisation. Issues such as interoperability and compatibility between systems may be particularly important at an organisational level.

Recommendations for organisations that produce evidence synthesis

It is important for evidence synthesis production organisations to foster collaborative, community-based environments that promote the development and use of AI systems, mindful of the need for accuracy and rigour. Clarity on organisational expectations for transparency and accuracy creates incentives for AI tool development teams to engage with users. It also provides some assurance that use of this tool will be acceptable within the organisation and drive adoption if it meets these requirements; thereby, justifying the investment necessary to develop a new tool to expected standards.

The OECD AI Principles are helpful for evidence synthesis production organisations in developing institutional policies about AI in evidence synthesis.⁸ The following recommendations aim to facilitate responsible AI use for organisations that produce evidence synthesis³.

Ensure best practice standards for responsible AI use are clear and integrated in your policies and guidelines:

- 4.1 Establish standards for responsible AI use, including rigour and transparency, that evolve with the changing AI landscape. This should include guidance for justifying and validating AI use in specific synthesis and procedures to address their misuse.
- 4.2 Provide specific requirements for accuracy and transparency so that system developers are aware of your expectations (see *AI tool assessment considerations in RAISE 3*).
- 4.3 Where possible, report organisational stance on use of AI tools, including details on the assessment and any caveats that individuals should be aware of before use.
- 4.4 Integrate AI governance in evidence synthesis with organisational governance on research integrity.

Promote, guide and support responsible AI use in your evidence synthesis activities:

- 4.5 Facilitate the development of training materials to support selecting and using AI tools in evidence synthesis.

³ See also EC Living guidelines on the responsible use of generative AI in research ¹¹

- 4.6 Provide support and guidelines to facilitate compliance with ethical, legal and regulatory requirements (see *ethical, legal and regulatory considerations when using an AI tool* in RAISE 3).

Monitor the development and use of AI within your organisation:

- 4.7 Monitor how AI is used in evidence synthesis practices across your organisation, and share experiences on effective use and the challenges, both internally and externally, to help improve practices over time.

Contribute to the ecosystem to help all roles continue to develop and grow, for example:

- 4.8 Collaborate with those across the ecosystem to explore and define standards and expectations that can be applied globally.
- 4.9 Engage in training (including within your organisation and evidence synthesis networks) and commit to ongoing learning about the responsible use of AI as it evolves, and remain informed about best practices.
- 4.10 Engage with AI system developers to create reliable systems. Whenever possible, and in accordance with data reuse policies and copyright licenses, provide data to help AI system developers test and validate their prototypes using real-world data. Where necessary, consider limited rights along with appropriate restrictions and controls, such as additional safeguards to protect content from unauthorised use through technology, terms of use, and contractual restrictions.
- 4.11 When entering partnerships, confirm whether commercial vs academic or non-commercial use has associated costs for the AI system developers, and ensure the datasets are carefully curated to ensure high quality.
- 4.12 Participate in networks to learn and to share knowledge about responsible AI use in evidence synthesis.

5. Funders and commissioners of evidence synthesis and AI

Role

Funders and commissioners are responsible for the design and evaluation of research programs and resources. They usually have the authority to establish expectations that maximise value for interest holders. When funders possess less expertise in evidence synthesis methods than those they finance, they may seek external assurances that the work they consider funding will meet expected standards for evidence synthesis and the use of AI within the synthesis processes (i.e., those detailed in RAISE).

This guidance aims to support all relevant funders, including organisations whose primary function is distributing funds, such as research funders and charitable foundations, as well as those funding synthesis work that applies AI pertinent to their interests.

Expertise

- Mobilising resources to achieve specific purposes

- Considering theories of change and how different activities may contribute to a desired change in a particular context
- Funding call and program design
- Evaluating proposals and monitoring research programs
- Connecting different roles in the same space
- Considering and mitigating consequences (intended and unintended) of funding processes

What they need to know

Funders need to know what constitutes best and reliable practice in their domain, which tools and technologies are suitable, and what areas require further development and/or funding. They tend to finance individual projects, but often within the context of a broader set of goals or programs of work. Therefore, they are interested in AI and data approaches that can integrate different pieces of work for greater impact.

As new technologies transition from rarely used innovations to part of a widely used infrastructure, funders must recognise the benefits and risks, and balance this with the AI tool or system market share.

Recommendations for funders of evidence synthesis production

Ensure best practice standards for responsible AI use are clearly articulated and integrated into your funding programmes:

- 5.1 Ensure that best practice standards for responsible AI use are clearly articulated and integrated into your funding criteria, adopting shared standards where available.
- 5.2 Ensure funded projects and grantees adhere to best practice standards for responsible AI (see *Recommendations for evidence synthesists* and/or *Recommendations for evidence synthesis methodologists* in RAISE 1). This includes ensuring AI tools are validated (see RAISE 2) and that grantees adhere to national and international ethical, legal, and regulatory standards (see *ethical, legal and regulatory considerations when using an AI tool* in RAISE 3). Full release or publication of training and testing datasets should be a best practice.
- 5.3 Ensure full and transparent reporting of AI use in funded evidence synthesis products that showcases the use of AI to be justified. This should involve demonstrating that the tools are methodologically sound, that they do not undermine the trustworthiness or reliability of the synthesis or its conclusions, and that they are appropriate for the specific evidence synthesis.
- 5.4 Mandate open science practices to support reusability, following the UNESCO Recommendation on Open Science ²³, which includes open publication, open data, open technology including open source, and open education.

Support the sustainability and generalizability of the products you support:

- 5.5 Recognise that evidence synthesis projects that AI and automation may lead to improved process efficiencies.
- 5.6 Ensure the complete, transparent and public reporting of all evaluations.

Contribute to the ecosystem to help all roles continue to develop and grow, for example:

- 5.7 Collaborate with those across the ecosystem to explore and define standards and expectations that can be applied globally.
- 5.8 Support the evidence synthesis through training on responsible and ethical AI.
- 5.9 Participate in networks to learn and to share knowledge about responsible AI use in evidence synthesis.

Recommendations for funders of evidence synthesis technology

Ensure best practice standards for responsible AI use are clearly articulated and integrated into your funding programmes:

- 5.1 Ensure that best practice standards for responsible AI use are clearly articulated and integrated into your funding criteria, adopting shared standards where available.
- 5.2 Ensure that funded projects and grantees adhere to best-practice standards for responsible AI and the principles of reliable evidence synthesis (see the *Recommendations for AI tool development teams* in RAISE 1 and RAISE 2). This includes ensuring AI tools are validated and adhering to national and international ethical, legal, and regulatory standards to ensure the AI systems are safe, secure and resilient.^{8,10} Full release or publication of training and testing datasets should be a best practice.
- 5.3 Fund the development of human-centred AI systems that emphasise human oversight. In this context, ‘human-centred’ means that humans should maintain control over the AI system while having agency and oversight of its operation.
- 5.4 Fund collaborative teams that involve and engage key interest holders in the development of AI systems and /or data governance frameworks (in line with the RAISE ecosystem vision), recognising that high-quality AI work typically requires collaboration between domain experts and technology experts.
- 5.5 Fund AI systems that consider responsible handover guidelines when transitioning an AI system from development to adoption (such as those provided by Sense About Science¹⁸).
- 5.6 Support independent, open evaluations that help produce community guidelines for the responsible use of AI technologies. Ensure the complete, transparent and public reporting of all evaluations.
- 5.7 Only support open-source projects and prioritise tools that promote open science methods, such as defaulting to open data standards whenever possible.

Support sustainability and generalizability of the products you support:

- 5.8 Ensure grantees make the details about the AI systems public and transparent (see *AI tool assessment considerations* in RAISE 3 for an overview of what evidence synthesists need to know about an AI system), including strengths, limitations, biases and how they might be mitigated, generalizability, funding sources and any interests. Ensure clear user guidance is available with clear terms and conditions.

- 5.9 Fund AI systems that do not require users to relinquish rights over or share their data for future system development and evaluation. Ensure your expectations align with existing and emerging standards for regulation and use.
- 5.10 Fund AI tools that commit to continued learning, development and monitoring of the AI system, with accessible and transparent appeals or feedback pathways for when systems do not perform as expected.
- 5.11 Fund generalist tools that can be applied to evidence synthesis from across different fields, as they often yield the best results in specific domains. When funding domain-specific tools is warranted, consider how they can be generalised.

Contribute to the ecosystem to help all roles continue to develop and grow, for example:

- 5.12 Collaborate with those across the ecosystem to explore and define standards and expectations that can be applied globally.
- 5.13 Support the evidence synthesis community in transitioning to the next generation of technology by funding training on responsible and ethical AI.
- 5.14 Participate in networks to learn and to share knowledge about responsible AI use in evidence synthesis.

6. Publishers of evidence synthesis

Role

Those who publish evidence syntheses are responsible for ensuring adherence to standards and policies regarding AI in conduct and reporting (i.e., those detailed in RAISE). This group includes publishers, journal editors, peer reviewers and organisations with editorial or quality assurance processes that disseminate evidence synthesis.

Expertise

- Publication ethics and research integrity best practices and standards.
- Editorial and publication policies, best practices, and platforms
- Reporting standards
- Societal relevance
- Copyright and data accessibility

What they need to know

Individuals involved in editorial decisions and the publication process of evidence synthesis submissions should be familiar with best practices in evidence synthesis, including the standards for conducting and reporting. They need to understand the principles of responsible use of AI and how its use should be reported.

Recommendations for publishers

The following recommendations are intended to promote responsible AI use among publishers of evidence synthesis, including publishing companies, journal editors, peer reviewers, and organisations that have editorial or quality assurance processes for sharing evidence synthesis. They

are also applicable to research on AI tools for evidence synthesis, including evaluations, which should adhere to the reporting framework detailed in RAISE 2 for reporting on the building and evaluation of an AI tool.

A growing number of publishing bodies have released or are consulting on recommendations for AI use in scholarly communication, such as STM⁹ and the European Association for Science Editors (EASE)²⁴. This guidance aims to align with these and build on additional evidence synthesis-specific considerations.

Ensure best practice standards for responsible AI use are clear and integrated in your policies and guidelines for authors and peer reviewers:

- 6.1 Establish standards for responsible AI use that incorporate rigour and transparency, adapting to the continually evolving AI landscape, and include procedures to address potential misuse.
- 6.2 Implement stringent quality assurance processes to ensure the methodological soundness and scientific rigour of evidence synthesis products that use AI. When applicable, share feedback with the evidence synthesists about their use of AI, particularly in situations where you question whether the model functioned as expected, if biases are revealed, or if you identify conditions in which the system is less reliable.
- 6.3 Integrate governance of AI use research submission with organisational governance about research integrity.
- 6.4 In line with expectations of open science, encourage evidence synthesists to share their data openly to aid transparency and to support further AI tool development.⁴

Request transparency and honesty from authors on their use of AI in their evidence synthesis:

- 6.5 Encourage authors to offer detailed descriptions of their research methods, including data sources, algorithms, and validation procedures that conform to expected reporting standards (e.g., PRISMA).
- 6.6 Uphold the importance of honest and accurate reporting of AI research outcomes, avoiding sensationalism and unfounded claims.
- 6.7 Require authors to report the limitations and uncertainties of their AI models and findings accurately, including the generalisability of the model to a given application.
- 6.8 Require authors to disclose any potential conflicts of interest related to their AI research.

Contribute to the ecosystem to help all roles continue to develop and grow, for example:

- 6.9 Collaborate with those across the ecosystem to explore and define standards and expectations that can be applied globally.
- 6.10 Engage in training and commit to ongoing learning about the responsible use of AI as it evolves, make sure you remain informed about best practices, and educate authors in responsible AI use.

⁴ Please note that, where appropriate, the sharing of data needs to respect the licenses under which it was obtained, and the imperative to share data openly does not override the need to comply with publishers' binding terms and conditions.

- 6.11 Promote open science practices, including access to research findings, datasets, and source code, where ethical and legal considerations permit, to allow for independent verification and replication.
- 6.12 Participate in networks to learn and to share knowledge about responsible AI use in evidence synthesis.

7. Users of evidence synthesis

Role

The evidence synthesis user will use the knowledge obtained from evidence synthesis outputs to make informed decisions. Users generally fall into four categories

- those who use the knowledge to implement a specific practice or intervention;
- those who broker the knowledge, including policymakers, guideline developers, or individuals making improvements to the research process;
- those who utilise or build upon the research, such as those advancing further research in particular areas of promise and/or need; and
- those who apply the research in educational settings regarding best practices in the field.

Additionally, it is increasingly recognised across all fields that research best practices should involve public members, so this group should also be considered a user of evidence synthesis.^{25,26} They are responsible for their decision to apply the knowledge.

Expertise

- Assessing the transparency and credibility of evidence synthesis products
- Relevant substantive domain expertise (e.g., health, social, environmental area)
- The applications of evidence synthesis findings (e.g., policy, implementation, future research directions, etc.)

Users are not expected to have expertise in the technical details of how AI tools used in evidence synthesis outputs work.

What evidence synthesis users need to know

Users of evidence synthesis should be able to determine whether the evidence syntheses are credible and have a reasonable degree of confidence that any AI tools used in conducting or reporting the evidence synthesis are accountable, transparent, valid and reliable. They should understand that the factors underpinning this confidence are influenced by the evidence synthesists assuring users that AI use was conducted with human oversight, by communicating the application of AI tools transparently, and by showcasing that the standards for responsible AI use in the evidence synthesis were adhered to (i.e., those detailed in RAISE).

Recommendations for users of evidence synthesis

Appraise the AI use in evidence synthesis before use:

1. Recognise that the growing use of technologies in evidence synthesis will likely expand the quantity and scope of synthesised outputs available for use.
2. Carefully consider the potential impacts of AI on any evidence synthesis outputs you intend to use and whether the producers of the syntheses were transparent about the accuracy and relevance of AI tools applied (the strengths and limitations of the use of AI should be included in the Discussion section of an evidence synthesis article, as well as alongside any AI use disclosures).
3. Recognise whether AI was used with human oversight; human oversight (or 'human in the loop') is fundamental to ensuring trust in the synthesis, its findings and conclusions, as it showcases that humans control the AI system and maintain agency and oversight over its operation.
4. If possible, consider the potential assumptions and biases of the AI tool(s) and its impact on the findings and conclusions of the synthesis (e.g., if AI is used to aid in study retrieval, may this inflate citation or publication biases?).
5. Distinguish between rigorous, high-quality evidence syntheses, where teams of human experts have conducted a review of the existing evidence with the help of AI tools, versus AI evidence discovery platforms and auto-generated summaries.
6. Carefully consider the journal or platform on which the evidence synthesis is published and whether they have adequate review and quality assurance processes.

Use the evidence synthesis in a responsible way:

7. When creating policy briefs or summary documents based on the conclusions of evidence syntheses, ensure that the potential impacts of AI tool use are clearly communicated.

Contribute to the ecosystem to help all roles continue to develop and grow, for example:

8. Collaborate with those across the ecosystem to explore and define standards and expectations that can be applied globally.
9. Engage in training, commit to ongoing learning about the responsible use of AI as it evolves, and stay informed about best practices.
10. Promote open science practices, including full and transparent reporting, access to research findings, datasets, and source code, where ethical and legal considerations permit, to allow for independent verification and replication.
11. Participate in networks to learn and to share knowledge about responsible use of AI in evidence synthesis.

8. Trainers of evidence synthesis methods

Role

Trainers of evidence synthesis methods hold a critical responsibility for building capacity in two important areas: 1) guiding evidence synthesists in the responsible use and reporting of AI tools; and 2) instructing evidence synthesis methodologists on how to conduct evaluations. Additionally, some trainers may also engage with AI tool development teams to communicate the needs and expectations of evidence synthesists.

Expertise

- Evidence synthesis and evaluation methodology
- How AI tools work, and the best ways of communicating this to different audiences
- Teaching and the ability to strategically capitalise on opportunities for capacity building

What trainers of evidence synthesis methods need to know

Trainers of evidence synthesis methods must understand how AI tools function and what other members of the ecosystem require to perform their roles effectively. They should also be familiar with how to evaluate AI tools, identify common evaluation mistakes, and contribute to establishing a cumulative evidence base (see *Building and Validating an AI tool* and *Conducting Evaluations to Build a Cumulative Evidence Base* in RAISE 2). Finally, they need to recognise when and how to use AI tools, when further validation is necessary, and when using a tool might be better avoided.

Recommendations for trainers of evidence synthesis methods

Ensure best practice standards for responsible AI use are embedded within your training materials

1. Introduce important concepts for evaluating AI tools and the commonly used performance metrics (see *conducting evaluations to build a cumulative evidence base* and *performance metric considerations for AI tool evaluations* in RAISE 2).
2. Build capacity among evidence synthesis methodologists to conduct rigorous evaluations.
3. Critically consider the accuracy and potential risks of any AI tool before inclusion in training materials.
4. If AI tools are included, aim to ensure trainees are equipped with the knowledge they need to know if they *should* use them (compared to if they *could* use them), for example, citing evaluations, information about understanding applicability, limitations, generalisability, and ethical, legal and regulatory considerations (see RAISE 3).
5. Emphasise the importance of human oversight (or ‘human-centred’ AI systems, indicating that humans should control the AI system and maintain agency and oversight over its operation), throughout the evidence synthesis process, as well as the issues that can arise when AI systems are applied inappropriately.

Contribute to the ecosystem to help all roles continue to develop and grow, for example:

6. Collaborate with those across the ecosystem to explore and define standards and expectations that can be applied globally.
7. Where appropriate, engage with evidence synthesists, organisations that produce evidence synthesis, and AI tool development teams to share learnings.
8. Engage in training, commit to ongoing learning about the responsible use of AI as it evolves, and stay informed about best practices.
9. Participate in networks to learn and to share knowledge about responsible AI use in evidence synthesis.
10. Actively seek opportunities to engage with and train evidence synthesis users because the ability to understand strengths and limitations of an evidence synthesis product, including the role of AI tools, is critical to informed use. Given the potentially large

impact of knowledge brokers, such as policy makers and the public, a clear understanding of limitations is particularly significant.

Conclusions

Evidence synthesis has been built on the principles of rigor, transparency and replicability, and these principles serve as cornerstones in evidence-based decision-making. However, it has become time-consuming and expensive, making it difficult to keep up with the demand of funders and users. While AI holds great promise to improve the timeliness and cost-effectiveness of evidence synthesis, it also presents challenges. At the time of writing, many AI solutions would undermine the principles on which evidence synthesis was built, risking a deluge of unreliable, biased, and low-quality systematic reviews and syntheses lacking essential transparency.

RAISE began in response to calls for guidance on the responsible use of AI in evidence synthesis. This has been challenging due to the relatively undeveloped and rapidly changing nature of the field. To better ensure more reliable use of AI tools, it is necessary to cultivate an ecosystem that supports the appropriate development, validation, and adoption of fit-for-purpose tools. RAISE encourages every individual and organisation from across the evidence synthesis ecosystem to understand how they should both contribute to and adapt responsible AI use to make evidence synthesis more sustainable. As AI and automation become embedded within evidence synthesis processes, it is vital to ensure that the core principles on which evidence synthesis was built are not compromised as the field evolves.

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